

Carbon footprint in denim manufacturing

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6.1 Introduction

‘Eco-chic’, ‘environmentally conscious’, ‘ethical consumerism’, ‘sustainable fashion’ and ‘clothing with a conscience’ are fashion buzzwords. Even more familiar, ‘natural’, ‘organic’ and ‘green’ are adjectives intended to shape consumers’ minds and purchases regarding more sustainable choices. Sustainability is a growing phenomenon in the world of fashion. Globalization has provided the necessary impetus for the apparel industry to produce and deliver fashion at a rapid rate [1–3]. Also, stringent enforcement of environmental laws by regulatory authorities shows that there is growing recognition regarding moving toward a more sustainable model in the textile and clothing industry.

Climate change, also known as global warming, is resulting in large-scale irreversible changes globally. A major promoter of this is the emission of greenhouse gases (GHGs), which are caused by the production and consumption of fuels and other manufacturing materials. The greenhouse effect is responsible for keeping the earth warm. High concentrations of GHGs affect the earth’s equilibrium, putting it out of balance, which will lead to extreme variations in weather. Unchecked burning of fossil fuels continuously releases carbon dioxide (CO₂) into the atmosphere [2]. The greenhouse effect is caused by GHGs that trap heat in the earth’s atmosphere. Uncontrolled global warming could influence global eco-regions. A rise in temperature globally will influence ecosystems, which could lead to several species being forced out of their territory because of changing conditions while others flourish [4–10]. Apart from these major drawbacks, minor effects of global warming are a rise in sea level, changes in climatic conditions and lessened snow cover, which may influence the ecosystem as well as human activities. Burning of fossil fuels to generate energy could increase the carbon footprint into the atmosphere.

The carbon cycle is an important cycle of the earth, and human activity is significantly influencing the various physical and chemical activities responsible for the earth’s climate [11]. Atmospheric concentrations of CO₂ have been increasing for decades, creating profound implications for future climate change, the productivity of agriculture and the function of ecosystems. Much research has demonstrated that global warming mainly results from the increasing emissions of GHGs as a result of human activities. The major contributing factor to global warming is CO₂, which accounts for approximately 80% of emissions from developed economies. To keep

track of one's GHG emissions is to be aware of one's carbon footprint. Carbon footprints are a measure of the impact that human activity has on the environment in terms of the quantity of CO₂ equivalents; it is measured in kilograms or tons.

The carbon footprint assessment (CFP) is an important approach for the control and management of GHG emissions. The method is based on the life-cycle concept and is used to assess GHGs released during the manufacture of goods and services, either throughout their entire life cycle or within specified temporal or spatial boundaries. The carbon footprint is the total amount of CO₂ and other GHGs released during the full life cycle of a process or product. It is calculated for the period of 1 year and is expressed in terms of the amount of CO₂ or its equivalent of other GHGs released [12]. Other kinds of GHGs which could be released are ozone and methane [13,14]. These GHGs are also considered for the evaluation of the carbon footprint but they are normally converted into equivalent CO₂ (i.e., the amount of CO₂ that could lead to a similar influence on global warming).

The major idea behind CFP originates from life-cycle analysis (LCA). As per the International Organization for Standardization (ISO) 14040 series, an LCA involves a methodical set of measures for compiling and investigating the inputs and outputs of products and energy and the related environmental impacts directly related to the functioning of a product through its entire life cycle, and within a given set of boundary conditions [15]. When accounting for CO₂ (or other GHG) emissions along the life cycle of a product, the term 'carbon footprint' is used. The idea behind 'footprint' comes from an ecological footprint accounting system, in which 'ecological footprint' is used as an indicator to measure the consumption of resources and waste assimilation necessities based on the economy or human population by means of a productive land area [16].

Denim is one the most popular types of clothing in the world; millions of pairs are made and sold in many different kinds of styles and colors. Having a pair of jeans has become the most personal thing a person can wear because it is said that it molds to the body. The reasons why jeans have become a neutral garment is that they can be worn in different ways, their composition is simple and they have changed with time. Denim production is a major subsector in the textile industry. Denim products are some of the most highly used in textile clothing, with continuous fashion and consumer preference, especially by young people. A survey by Cotton, Inc. showed that the global denim jeans market was projected to grow by 8%, from \$55 billion in 2015 to \$59 billion by 2021, with Latin America and Asia expected to lead the increase.

Denim manufacturing and other textile-based industries continue to be a source of specific environmental pollution because of considerable amounts of mineral substances that are used to process textile raw materials and semifinished products. The life cycle of a pair of jeans is divided into two categories: the manufacturing stage and the use of the jeans until the end of its cycle. The production stage is responsible for approximately 59% of the climate change impact, whereas the use of the jeans and end-of-life process accounts for the remaining 41%. This causes jeans to be responsible for about 2 kg of CO₂ per functional unit.

The first part of this chapter provides the basis for the carbon footprint and its estimation in the denim manufacturing process. Furthermore, the carbon footprint in various stages of denim manufacture, such as cotton production, denim yarn manufacture, fabric production, garment manufacture, transportation and distribution, are dealt

with in detail. In the second part of the chapter, the carbon footprint of denim processing, such as dyeing and washing, and the carbon footprint during the consumer user phase are elaborated upon. In addition, strategies to reduce the denim fabric carbon footprint, such as 3R concept and other energy efficient technologies and processes, are briefly discussed.

6.2 Carbon footprint standards and methods of calculation

The major objective behind the application of standards is to ensure easy application in practice [17]. ISO standards for LCA, which has been in existence for about 3 decades, have had a vital role in shaping carbon footprint standards: ISO 14040 for LCA and ISO 14064 for GHGs. Yet another standard, ISO 14067, is under development [18]. Another development is the PAS 2050, by the British Standards Institute, which covers a wide range of diverse products based on the LCA method. It is the most detailed and comprehensive product carbon footprint methodology available [17,19]. However, one of the main issues in the PAS 2050 methodology is the varying results arising owing to the freedom given to use any database. For example, one product within a specific system boundary can have two different values depending on the databases used for calculations, because the standard does not require the use of specific databases. In the absence of primary data, different values are often used from various databases [20].

Other standards in wide use are the GHG Protocol of the World resources Institute and the World Business Council for Sustainable Development and their Product Life Cycle Accounting and Reporting Standard [18], and the Basic Guideline of the Carbon Footprint of Products of the Japanese Industrial Standards Committee.

Methodologically, there are three ways to carry out a life cycle-based CFP analysis, depending on the scale of application:

1. a process- or product-based analysis (or bottom-up) approach which accounts for the environmental impacts of individual products from cradle to grave or only on-site. In this method, mostly first-order, and some second-order impacts are considered [21]. This also depends on the question of how to select boundaries. If used to make carbon footprint comparisons of various products, it is important to mention the boundaries used because it can cause confusion or errors. Furthermore, if applied within a larger context such as a sector (industry, household or government), estimates are based on assumptions and therefore could be far from reality;
2. an environmental input–output analysis (or top-down) approach can be used in a broader context. Input–output analysis in economic accounts provides a picture of all economic activities at the sector level. A detailed understanding of micro-level impacts is hidden or limited owing to aggregation of data and homogeneity of the system required by this approach [20,22];
3. combining the strengths of these two approaches, a hybrid approach is proposed in which the process-based and input–output methodologies are integrated. However, hybrid modeling is rarely practiced because of the high methodological and data sophistication required [18,23].

6.3 Carbon footprint in textile and clothing sector

Sustainable fashion has been around since the early 1600s, as a way of life rather than a choice. Before the Industrial Revolution, people conserved resources because of the high costs of labour involved in the conversion which are basically meant for the wealthy people [24–26]. Raw materials (textiles) for apparel came from nature and required a long time to be transformed into fabrics. The labour required to produce a high-quality product contributed to the high cost of fabrics and apparel. Hence, quality fabrics were worn by wealthy people as a way to express their wealth in society. By the end of the 18th century, because of the Industrial Revolution, several types of machinery were introduced to spin yarn and weave fabrics, and the textile industry flourished. Synthetic dyes, manufactured fibres, ready-to-wear clothes, fashion magazines, etc., were products of the Industrial Revolution that caused the textile sector to mature [27].

The textile and clothing industry has created a lot of material wealth for humans but it also consumes a lot of energy, which causes massive emissions of CO₂ and other GHGs. To protect the environment for human survival, the industry should develop a low-carbon way to achieve sustainable development [28,29]. Because of the long and complicated nature of the supply chain, the textile and clothing sector is responsible for creating a significant carbon footprint and is a major source of GHGs. An environmental impact occurs at each stage in the life cycle of garments, from the cultivation of fibres or the production of synthetic fibres to the various manufacturing sequences involved in converting fibres to fabric and the dyeing and finishing of garments for delivery to consumers. Disposal of products after use and the discharge of effluents also cause significant impacts [30]. Cheap clothing has become a disposable product and most European consumers recycle it, donate it or dispose of it; the remainder goes to landfills. Regardless of the economic downfall globally, increasingly more garments, home textile materials and other textiles are purchased by the consumers annually.

The problem is more severe in the developing world, where textile production contributes to a larger percentage of the gross domestic product and processing facilities are out of date. The carbon footprint of a textile product varies depending on the fibre used and the method of processing. The carbon footprint of any product can be examined by the ‘level of embedded energy’ within the product. This is the net amount of energy consumed at each stage of the production process. Despite stringent environmental laws and regulations, the textile industry cannot meet these requirements satisfactorily. Several reasons lie behind the huge impact of textiles on the environment:

- The vast majority of fibres produced are synthetic and petrochemical-based fibres such as nylon and polyester consume a huge amount of energy to manufacture. In addition, chemicals used during the manufacture of these materials end up as toxins polluting the air, soil and water.
- Traditional methods of cultivating cotton contribute heavily to global warming and are detrimental to the environment. Cotton cultivation necessitates enormous quantities of water and energy and the intensive use of chemical fertilizers and pesticides.
- Bleaching and coloring of fabrics also warrants an enormous quantity of water, dyes, chemicals and energy. Around 1 million tons of chemicals and dyes is used every year to color fabrics.
- The wet finishing process uses a huge amount of water and energy.

The textile industry is a huge industry and a massive polluter. It uses two things to a great extent: water and chemicals. Wet processing of textiles involves desizing, scouring, mercerizing, dyeing, printing, finishing, etc., and requires a lot of chemicals to be applied on fibres or yarns for fabrics. Water is required at almost all stages of fabric processing. It is used as a solvent to dissolve chemicals and for washing between the various sequences of wet processing. The bleaching operation releases an organo-chlorine compound which is harmful to the environment. At least 2000 types of chemicals are employed by this sector [29,31].

6.4 Denim

Among various textile products, denim has received wider acceptance compared with other fabrics. It is worn by various age groups and classes and both genders. Denim fabric is a hard and durable 3/1 warp-faced twill cotton fabric with indigo-dyed warp yarn and white color filling yarns, with a weight of 14.5 ounces/square yard. As long ago as the 17th century, denim was used to make trousers, upholstery and awnings [32].

6.4.1 Definition of denim and jeans

- The Textile Institute defines denim in *Textile Terms and Definition* thus: ‘Denim is conventionally a 3/1 warp faced twill fabric made from yarn dyed warp and un-dyed weft yarn and Jean is a 2/1 warp faced twill fabric used chiefly for overalls or casual wear’.
- Based on the *Encyclopedia of Textile Finishing*, edited by Professor Dr Hans-Karl Rouette, published by Springer, ‘Denim is a firm 2/1 or 3/1 cotton twill-weave fabric (work wear twill) with a dyed blue warp and raw white weft. The warp is sometimes dyed during sizing’.
- ‘Jeans originally an American term derived from Genes (cloth from Genoa). The term was coined in the USA to describe tight-fitting trousers made of cotton twill, usually woven with a dark blue dyed warp and undyed weft, stitched on seams and pockets and having copper rivets here and there. As a rule, blue jeans are dyed with indigo and thus have limited color fastness to rubbing and washing’.
- Conventionally, denim means sturdy, indigo-dyed cotton fabric woven with a dyed warp yarn and natural fill yarn available in various weights and qualities that is used to make pants, skirts, jackets, shirts and shorts, etc.; jeans are pants made from denim.

6.4.2 Main features of denim fabrics

Denim fabrics are generally woven from coarse yarn and are dyed with indigo dyes in warp. They are hard wearing, high-density fabrics that are high gram per square meter. Denim is popular for long-term use and is a fashion fabric for casual wear. General features of the denim fabric are as follows [33]:

- cotton or cotton polyester blended, durable and heavy-weight twill-weave
- warp-faced twill fabric and colored warp yarns dominate on the face and the white filling yarns on the back side of fabric
- colored warp yarn and white filling yarn
- left-hand twill with a blue (indigo) warp, white weft for use in apparel in a variety of weights

- weight of the denim ranges from 203.46 to 474.74 g/m² or more in a 2/1 or 3/1 interlacing pattern
- may be napped or printed, made with spandex or other stretch yarns, or otherwise modified for fashion

6.4.3 Raw materials of denim fabrics

To manufacture a better denim fabric, processing conditions have to be optimal as far as the raw material is concerned. For raw cotton, the ring and rotor spun yarns made from it should have the following quality specifications [34]:

- minimum staple length, 2.7 cm
- short fibres (less than 0.5 inches) percentage, below 40%
- fibre fineness, 4.0–4.5 µg/inch
- yarn characteristics intended for use in denim manufacturing should at least conform to the Uster 25% standard
- common count range of denim warp yarns is 50–90Tex and weft yarns 75–120Tex; finer yarns as fine as 25Tex in twill or plain weave are repeatedly used in denim shirts
- yarn twist multiplier: 4.5–5.0 for warp yarns and 4.2 for weft yarns
- low yarn hairiness
- better yarn strength and uniformity

6.4.4 Classification of denim fabrics

Denim fabric can be classified based on several factors:

1. Based on weight/unit area:
 - a. light weight: 4.5–7 oz/square yard
 - b. medium weight: 7–10 oz/square yard
 - c. heavy weight: 11 and above oz/square yard
2. Based on colour:
 - a. indigo vat
 - b. sulphur black
 - c. pigment
3. Based on slub yarns
 - a. warp yarns are slub only
 - b. weft yarns are slub only
 - c. cross-hatch: both-ways slub, i.e., both warp and weft yarns are slub
4. Based on fabric structure/design:
 - a. plain
 - b. twill (S-twill, Z-twill, herringbone, or zigzag twill)
 - c. right-hand twill, diagonals run toward to the right
 - d. left-hand twill, diagonals run toward to the left

6.4.5 Manufacturing process of denim

Denim, in particular blue jeans, has the biggest market share in sales of textile products around the globe owing to its popularity in all countries and age groups. The denim production value chain is optimized for mass production and with production

facilities to meet the global demand. The higher demand and production of denim could lead to a significant influence on the environment [34]. Hence, technological developments in every step of denim production could provide cost savings to the industry and the environmental impact of denim. In evaluating the environmental impact of denim manufacture, it is vital to take a holistic approach by considering all of the activities involved in manufacturing denim, such as raw materials, manufacturing, transportation, use and disposal.

The denim manufacturing process is similar manufacturing grey fabric up to the weaving process, except that for denim fabric the warp is dyed at the sizing stage compared with dyeing at the finishing stage in normal fabric production [35]. Fig. 6.1 shows the entire textile value chain in the manufacture of denim fabric, from cotton fibre to consumer use. The process flowchart of denim fabric manufacturing is shown in Fig. 6.2, which involves spinning, warp preparation, fabric manufacturing and finishing processes.

6.4.5.1 Spinning

The first process in denim manufacturing is the spinning of yarn in either a ring spinning or rotor spinning system. The fibre to yarn conversion is composed of several processes such as opening and cleaning, carding, drawing, roving and spinning. For rotor spinning, the roving process can be eliminated. Opening and cleaning of fibres are carried out in a series of machines in a blow room. Carding individualizes the fibres and removes neps, short fibres and remaining trash, and produces a rope-like form called sliver. The uniformity and parallelization of fibres are improved through a drawing process. In rotor spinning, the drawn sliver is fed into the rotor machine to convert it into yarn. In ring spinning, the drawn

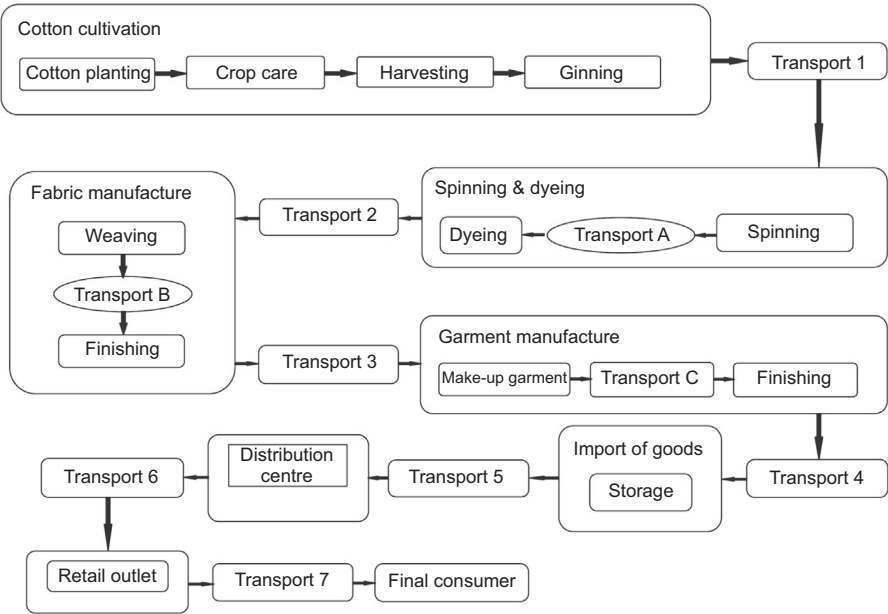


Figure 6.1 Textile value chain in denim manufacturing from cotton to consumer.

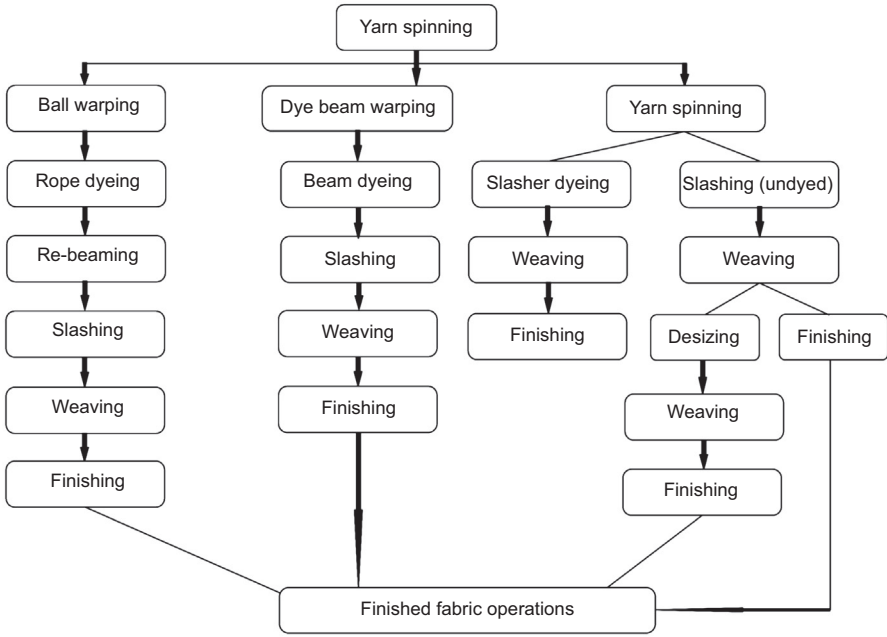


Figure 6.2 Process flowchart of denim fabric manufacturing.

sliver is further reduced to a roving form in a speed frame and then is converted into yarn in a ring frame [36–38]. Generally, denim fabric is 3/1 warp-faced twill fabric made from an indigo-dyed warp and an undyed weft yarn.

6.4.5.2 Warp preparation: dyeing and sizing processes

Dyeing and sizing of warp yarns are carried out in two methods:

- Yarn from several beams is combined to produce a warp sheet and then is dyed and sized on the same machine.
- A rope is made by combining about 350–400 yarns and about 12–14 ropes are passed near each other through the continuous dyeing machine. After dyeing, ropes are dried on drying cylinders and then a worker’s beam is prepared. After that, sizing is carried out by a conventional method.

In denim manufacturing, dyeing and sizing processes can be classified into three categories:

- continuous indigo-rope dyeing and sizing
- continuous indigo dyeing and sizing
- indigo–back beam dyeing and sizing

Continuous indigo-rope dyeing and sizing

Yarn from ring cops is converted into bigger packages as cone or cheese in cone winding and is positioned on the ball warper on which 350–400 yarns are formed into a

rope and are cross-wound into a ball shape according to the length of the warp beam. During this process, lease rods are inserted at frequent intervals as required for further processing on a long chain beamer. According to the size of the rope dyeing plant, 12–24 ropes can be dyed at a time [39–41]. The ropes are then opened on the long chain beamer through a tension roll and expansion comb, wound onto a back beam and then sized.

Continuous indigo dyeing and sizing

In this method, instead of ropes, back beams are processed on the dyeing/sizing machine. The warp yarn is indigo-dyed, oxidized, dried and sized in one continuous process. Although this method is less cumbersome compared with rope dyeing, the risk of individual thread breakage is greater than for rope dyeing [41,42].

Indigo–back beam dyeing and sizing

In this method, indigo dyeing and sizing are done in two different stages. In the first stage, back beams are indigo-dyed, oxidized, dried and wound onto a batch roll; in the second stage, the batch roll is sized, dried and wound onto a weaver's beam [43].

6.4.5.3 Weaving

The weaving process can be carried out on conventional weaving systems such as automatic looms or shuttle-less looms such as air jet, rapier, and projectile weaving machines. The slow speed of the conventional shuttle loom results in less production and extreme wear and tear to machines. Generally, denim fabrics are woven in shuttle-less weaving machines.

6.4.5.4 Finishing

The woven fabric wound in a roll form is inspected for defects on fabric inspection machines for immediate corrective actions. The denim fabric then goes through a range of finishing processes such as singeing, brushing, washing and drying. Singeing and brushing removes impurities from the fabric's surface and aids in making the fabric's surface even. The finished denim fabric is then sent for making-up, where the fabric is cut into the required width as per the size [44,45]. The finished denim fabric is then inspected for defects such as uneven dyeing, bleaching and dyeing defects, oil stains, patches, etc., and then is categorized according to the quality.

6.4.5.5 Garment manufacturing

Garment manufacturing is a process in which the two-dimensional fabric is transformed into a three-dimensional garment. Garment manufacturing involves several operations such as pattern making, spreading, cutting, sewing, pressing, inspecting and packaging. Accessories and trims are selected based on requirements and style aspects of the garment. Generally, the output from this process is the final product of the manufacturing sector, excluding some special effects conveyed on it owing to the requirements of functions or design concepts.

6.5 Carbon footprint and its estimation

The concept of a carbon footprint revolves around transportation and heating issues. The fabric necessitates a large CO₂ footprint. According to US information, the textile industry is the fifth largest contributor to the carbon footprint among many countries. The textile industry is large and produces a large amount of GHG. Textile production is estimated to produce around 60 billion kilogram of fabric. The total quantity of water and energy required to manufacture a fabric includes 6–9 trillion litres of water and 1074 billion KWh of electricity. The Bureau of International Recycling states that collecting 1 kg of used cloth could reduce around 3.6 kg of CO₂ emissions. Recycling textile products instead of disposing of them could reduce 6000 L of water, 200 g of pesticides and 300 g of fertilizer [46].

To evaluate the carbon footprint in fabric, the ‘embodied energy’ in the fabric has to be considered, i.e., the total energy used at each step of the process needed to create the particular fabric. To determine the embodied energy in the fabric, the energy used in two separate fabric production steps is determined [47]:

- The first step is to find out from what type of fibre the fabric is made, because that gives an idea about the energy needed to make the fibres used in the yarn. The type of fibres influences the carbon footprint of the fabric to an extent.
- Next, the energy needed to convert the yarn into fabric has to be taken into account. Once any material becomes a ‘yarn’ or ‘filament’, the amount of energy and conversion process to weave that yarn into a textile is consistent irrespective of the type of fibre.

6.5.1 Factors influencing the carbon footprint of the fabric

- **Use phase:** This segment of the life cycle of a product influences GHG emissions to a greater extent. The largest contributor to a garment’s life-cycle GHG footprint is laundering. Garments requiring the largest energy during washing, drying and ironing needs higher energy input; hence laundering itself accounts for 40%–80% of total life-cycle GHG emissions. Machine drying consumes the highest energy and produces higher GHG emissions in garments. On the other hand, hand washing and dry cleaning require lower energy, leading to lower GHG emissions.
- **Raw material:** The type of fibre has a vital role in GHG emissions. For example, synthetic fibres produce comparatively high GHG emission owing to higher energy requirements in fibre production compared with natural fibres. On the other hand, plant fibres such as cotton and linen have lower GHG emissions. Linen requires a relatively low amount of pesticides, fertilizers and irrigation compared with cotton, leading to much less GHG emissions. Animal fibres, especially wool, emits higher GHG owing to methane emissions from sheep. The type of washing, laundering and ironing process also depends on the type of fibres. For example, wool needs to be dry-cleaned or hand washed rather than machine washed and linen needs more ironing.
 - **Sourcing and manufacturing locations:** GHG emissions vary depending on the sourcing and manufacturing locations: for example, as a result of differing energy sources.
 - **Dyeing:** The dyeing process has little impact on the emissions of GHG.
 - **Garment makeup:** Garment assembly has a considerable impact on GHG emissions.
 - **Packaging:** According to many studies, packaging has a low GHG impact because packaging is often composed of recycled materials. Packaging generally depends on the type of garment.

- **Transportation:** The transportation of goods has a small portion of a garment's total carbon footprint. However, it depends on the distance products are transported and whether transportation is ship-based or air-based. GHG emissions could increase significantly if air transport is used in any stage of product manufacturing.
- **Garment lifespan:** The short life of clothing could increase total GHG emissions in ways not considered in a typical garment LCA. For example, a garment which is laundered only once before being discarded will have lower GHG emissions compared with that of a garment used and laundered more than 100 times.
- **Garment disposal:** GHG emissions resulting from the disposal of clothing are meager as per the LCA and are caused by the emission of a small quantity of methane during the decomposition of natural fibres. Certain disposal options reduce GHG emissions, however. The incineration of natural fibres in a waste-to-energy plant may shift the use of petroleum-based raw materials.

The carbon footprint in jeans manufacturing was determined by various denim manufacturing companies such as Levi Strauss & Co., Blue Dot and Crystal Denim to ensure that their jeans are sustainable. Levi Strauss & Co., one of the world's leading denim manufacturers, started a self-analysis program in 2006. They analysed the company's internal and external environmental impacts throughout their product line. In their LCA, the life cycle of jeans from 'cradle to grave' was reported as a collection of seven independent systems. Analysis started in the cotton production sector and then covered step-by-step denim material production such as fabric production, garment manufacturing, transportation and distribution, consumer use and recycling. Furthermore, the company listed the impact of the waste stream from the product manufacturing process in landfills. In their report, they mentioned that making one pair of denim jeans required almost 920 gallons of water and 400 MJ of energy and expelled 32 kg of CO₂. They also pointed out that the major quantity of water consumed (68%; 2565 L) was in fibre production. The percentage of water consumption is comparatively high: in all phases of garment production likely in fabric manufacturing, 6% (236 L); during sewing and finishing, 1% (34 L); in washing and packing, 2% (77 L); and at the customer's end, 23% (860 L). Through their carbon reduction programme, the Crystal group, a Hong Kong-based denim manufacturing group, reduced the carbon intensity 2.21–1.77 kg CO₂ per piece. Similarly Mud jeans went through several steps to reduce their carbon footprint in jeans manufacturing and reduced the carbon footprint of blue jeans from 23.45 to 5.13 kg CO₂e by streamlining the process and raw materials.

6.6 Carbon footprint in each stages of denim manufacturing

Levi Strauss & Co. classified the LCA of denim into seven different independent systems [48]:

1. cotton production
2. fabric production
3. garment manufacturing
4. transportation and distribution

- 5. consumer use
- 6. recycling, which then goes back to Step 2
- 7. waste stream in a landfill

The carbon footprint in the life cycle of denim with respect to the quantity and percentage is shown in Figs 6.3 and 6.4, respectively. The figures show that the consumer care phase (use phase) has the highest impact on climate change (driven by the high use of nonrenewable energy), followed by fabric manufacturing and transportation.

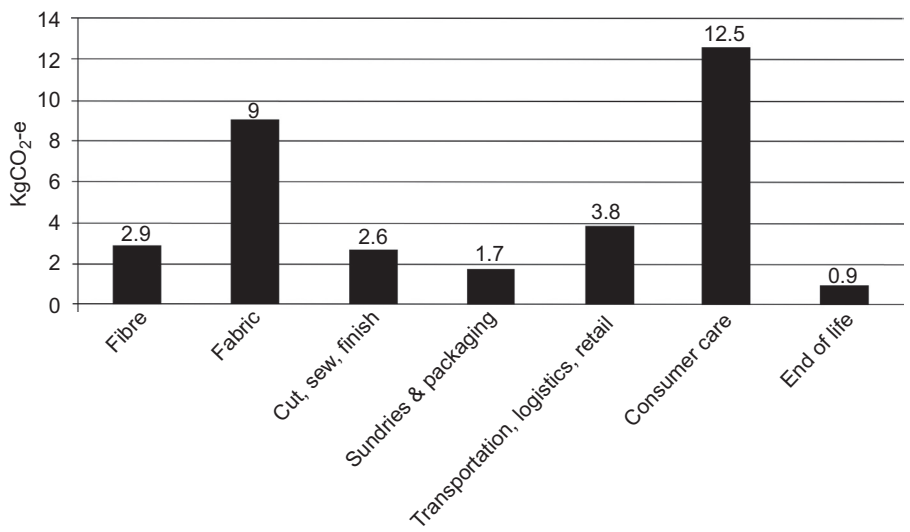


Figure 6.3 Climate change impact in the life cycle of denim jeans, in quantity. Levi Strauss & Co. Life cycle of a jean, reproduced with permission.

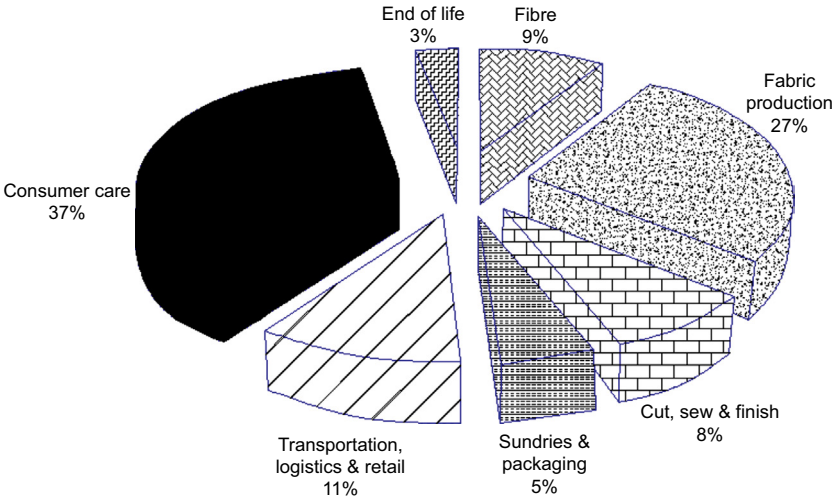


Figure 6.4 Climate change impact in life cycle of denim jeans, in percentage. Levi Strauss & Co. Life cycle of a jean, reproduced with permission.

6.6.1 Cotton cultivation and harvesting

The cultivation of natural fibres involves several factors such as field preparation, planting operations such as irrigation, weed control, pesticide, and the type of harvesting, which needs to be considered to reduce the carbon footprint. Cotton cultivation and harvesting are indispensable for denim manufacturing, although they are resource intensive and involve political and social aspects [49].

Cotton is considered one of the world's major crops; it contributes to about 3%–5% of the world's cultivatable surfaces. Cotton cultivation and harvesting are usually known for their unsustainable nature, owing to the use of pesticides, fertilizers and water. Cotton production can be both a contributor to and a 'victim' of climate change; it contributes around 0.3%–1% of total global GHG emissions. Current cotton-growing methods and practices are not considered sustainable because they damage water, soil and allied ecosystems, and contribute to high social costs and are a threat to regional economies, depending on cotton farming and associated textile industries.

To manufacture a pair of jeans, approximately 1.5 pounds of cotton is essential, which in turn requires about 1500 gallons of water. To cultivate and harvest the cotton, numerous stages of transport are necessary, and based on the number of steps each supplier can take in one facility, the quantity of carbon emissions will change. By taking into consideration the use of insecticides in cultivation, cotton growers use 25% of chemicals applied worldwide, including highly toxic organophosphates. Machines such as pesticide sprayers and farm equipment, which run on oil, require about a pound to harvest sufficient cotton to manufacture a single pair of jeans. Not only are the environmental impacts from these chemicals apparent within the industry, the social impact is compelling and sometimes is overlooked. To prepare the soil, harvest the cotton and transport cotton to the factories, a large amount of energy is used. In total energy consumption is around 60 MJ/kg [50,51].

The social impact of high-priced pesticides is horrible in India. In 2003 alone, more than 17,000 farmers committed suicide. Most were from cotton-growing areas where costly chemicals and genetically modified cotton seeds left them deeply in debt. The use of genetically modified cotton seeds could be an alternative method to conventional pesticide spraying, which is also revealed to be a different threat to social and environmental sustainability. Although India's share of CO₂ emissions is only 3.11%, the embodied energy of organic cotton from India is higher than conventionally produced cotton from the United States because the yields are much less in India, requiring more land to cultivate the same quantity of cotton. The energy distributions in cotton farming were studied by Nigam et al. and are shown in Fig. 6.5.

The figure shows that fertilizer consumes a major portion of the energy input along with diesel used for farming operations and electricity for irrigation. Fig. 6.6 shows GHG emissions from cotton farming. The study reveals that the cotton farming stage has 2.1076 CO₂eq/kg cotton yarn without allocation and 0.8430 CO₂eq/kg cotton yarn on a mass basis and 1.8336 CO₂eq/kg cotton yarn on an economy basis. From the figure can be seen that the major contribution is from the use of synthetic fertilizers. As much as 60%–70% of seed is available from seed cotton during ginning.

The workforce required to process 1 kg of cotton is calculated to be 0.002 workerh/kg. The electrical energy required to process 1 kg of yarn is 0.5863 MJ/kg. Based on the electrical energy consumed for ginning, the GHG emissions are calculated to be 0.1310 CO₂ eq/kg cotton yarn [52].

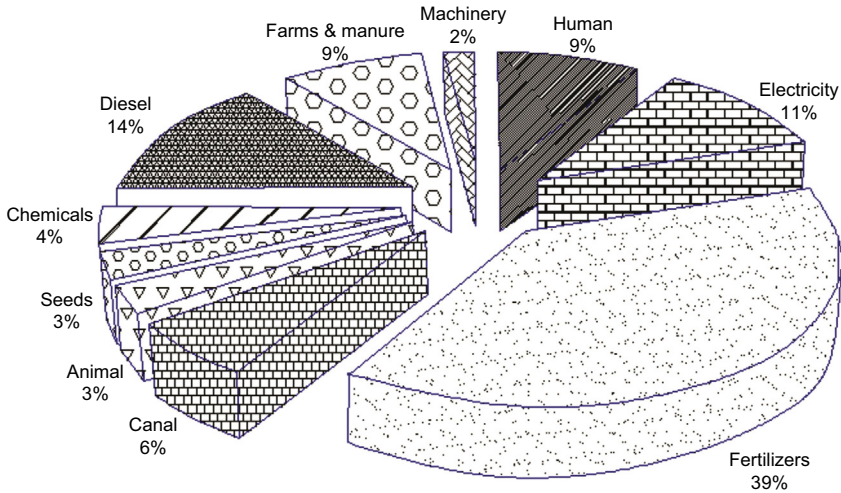


Figure 6.5 Energy distribution in cotton farming. CIAT, reproduced with permission. Clothing Industry Training Authority (CIAT), Apparel supply chain–carbon assessment about jeans producing. http://www.cita.org.hk/wp-content/uploads/2013/10/Fty_rpt_Eng_20130912.pdf.

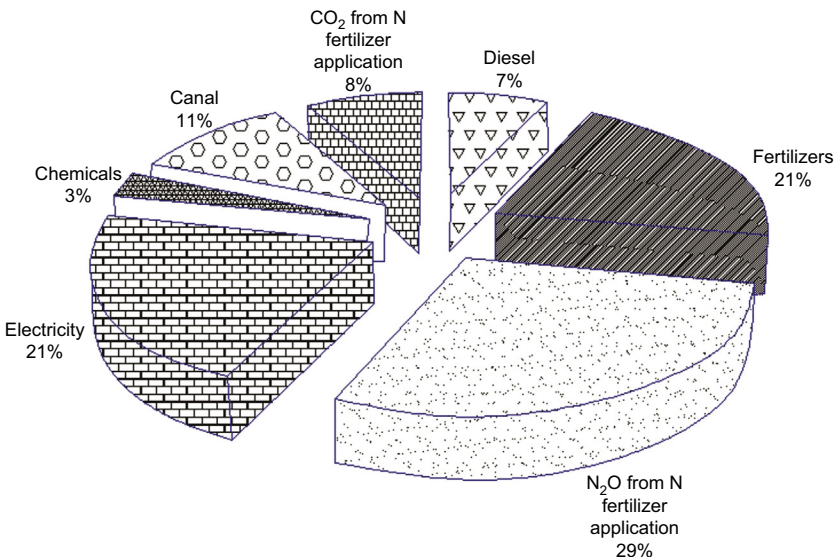


Figure 6.6 Greenhouse gas emissions from cotton farming. CIAT, reproduced with permission. Clothing Industry Training Authority (CIAT), Apparel supply chain–carbon assessment about jeans producing. http://www.cita.org.hk/wp-content/uploads/2013/10/Fty_rpt_Eng_20130912.pdf.

6.6.2 Cotton spinning

The value chain for cotton yarn production involves the following processes:

1. cotton cultivation and the harvesting phase, including fuel or energy-intensive material inputs such as fertilizers, herbicides, seeds, diesel fuel and electricity for irrigation, machinery and labour used for agriculture.
2. transportation to move agricultural products from the farm to the ginning site
3. ginning process, which consumes electricity
4. transportation to spinning industry
5. conversion of fibre to yarn, which uses mainly electricity and water for humidification
6. packaging

A process map for the manufacturing of carded ring yarn is shown in Fig. 6.7. Spinning starts with the blow room in which opening, cleaning, blending and dust removal are performed using beaters and grid bars. The cleaned tufts are then transferred to the carding machine through pipelines called a chute feed system. The fibres are then individualized, neps and short fibres are removed, remaining trash is cleaned and sliver is formed in carding. The drawing process improves the evenness and parallelization of fibres in sliver through doubling and drafting. Then the sliver is thinned to the roving stage in a speed frame, twisted and wound on a roving bobbin for the ring frame, where it is further thinned into yarn. In the case of rotor yarn, the slivers are directly converted into yarn.

The GHG emissions distribution study conducted by the Clothing Industry Training Authority in the cotton ring spinning industry is shown in Figs 6.8 and 6.9. The ring spinning process and ring frame machine contribute to a larger extent to GHG emissions and electricity consumption, followed by the winding process and machine. Incorporating energy-efficient processes and materials into these processing stages would offer a potential opportunity for carbon reduction [53]. Auxiliary production facilities used in the production process, such as air compressors, lighting, electric fans, humidification systems, and automatic waste extraction, have a vital role in energy consumption and hence a potential opportunity to reduce carbon emissions.

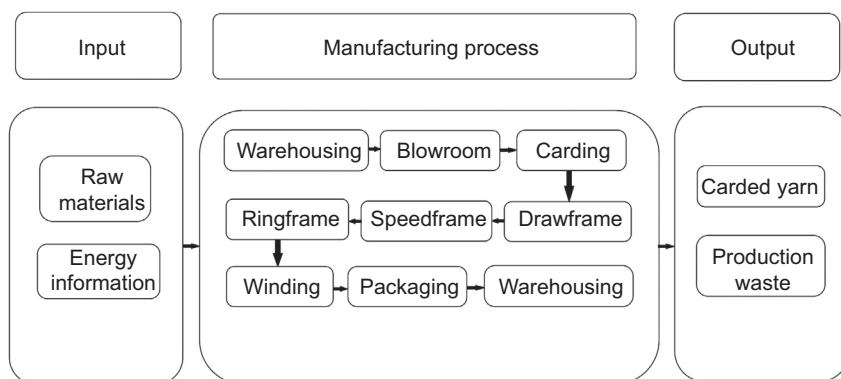


Figure 6.7 Process flowchart of carded yarn.

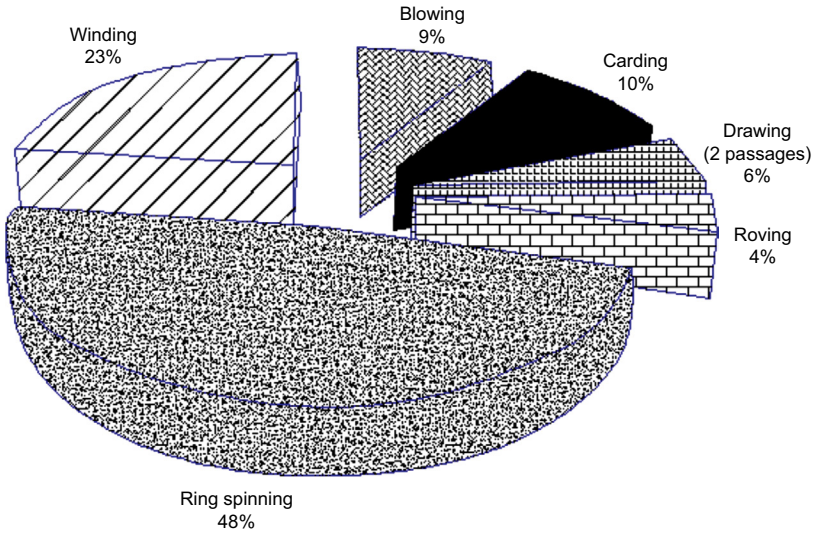


Figure 6.8 Greenhouse gas emission distribution in spinning, by process. CIAT, reproduced with permission. Clothing Industry Training Authority (CIAT), Apparel supply chain–carbon assessment about jeans producing. http://www.cita.org.hk/wp-content/uploads/2013/10/Fty_rpt_Eng_20130912.pdf.

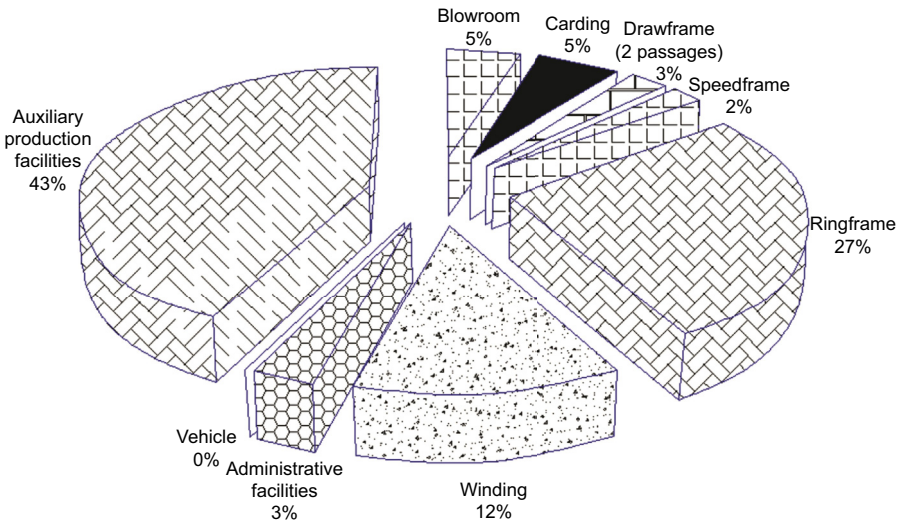


Figure 6.9 Greenhouse gas emission distribution in spinning, by emission source.

6.6.3 Dyeing and sizing

Rope dyeing is most commonly used in denim manufacturing because of the better uniformity achieved in this method compared with slasher dyeing. It produces a ring-dyeing effect in yarn, i.e., colour on the sheath of the cotton yarn without colour (white) on the

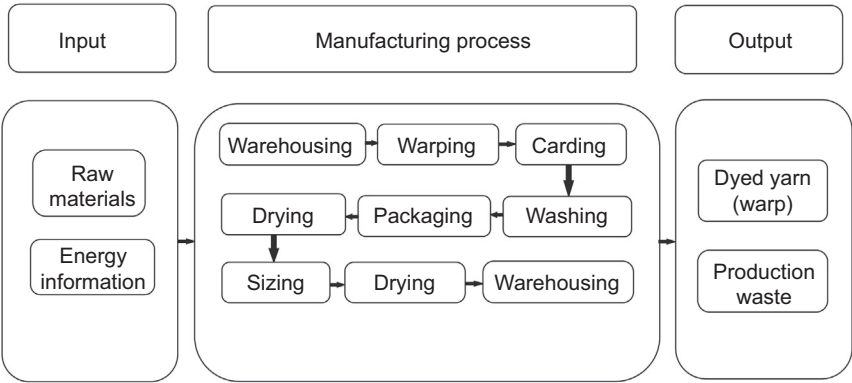


Figure 6.10 Process map of yarn dyeing.

core of the yarn, because it forms a coating on the outer layers of the yarn and fibre during the dyeing process. The most commonly used dye for denim is indigo [54]. This type of dye is known as a vat dye; it is insoluble in water and has less affinity toward cellulose fibres. Hence, indigo dyes are converted into leuco form, which forms a coating on the outer layer of yarn. The indigo is then made insoluble through oxidation by exposing the yarn to air. During sizing, a size add-on of 8%–12% is applied over the warp thread to improve the strength and frictional resistance of the warp yarn [55,56]. A process map for dyeing and sizing for denim manufacturing is shown in Fig. 6.10.

An average pair of jeans has around three-quarters of a pound of chemicals in it. Although natural indigo dyes can be produced, most indigo dyes are synthetically produced. The distribution of GHG emissions in cotton dyeing and sizing, process by process, and emission sources during the denim process are shown in Figs 6.11 and 6.12, respectively. Fig. 6.11 shows that the dyeing and sizing processes themselves consume around 95% of total energy. Analysing the energy source for each process and the behaviour of machines during the process may provide a possibility of reducing the carbon footprint in these processes [57]. Fig. 6.12 shows that steam boilers are major consumers of energy, about 80%, followed by dyeing and sizing machines.

Any kind of dye is basically a polluting process. Textile processing units need a nearby water source to break down dyes and to wash and treat the fabrics. Synthetic indigo dyes are normally sulphur based and have serious detrimental effects for both the people that work with them and the natural environments where they are discharged [58]. In addition to sulphur-based dyes, sodium hydroxide, hydrosulphate and formaldehyde are used to manufacture synthetic dyes. Even if the color is removed from wastewater, this does not necessarily mean that the water is clean: Sulphur dye residues tend to remain in wastewater even after treatment. Use of natural indigo dyes could minimize the carbon footprint level.

6.6.4 Denim fabric weaving and finishing

A process map of denim fabric weaving and finishing is shown in Fig. 6.13.

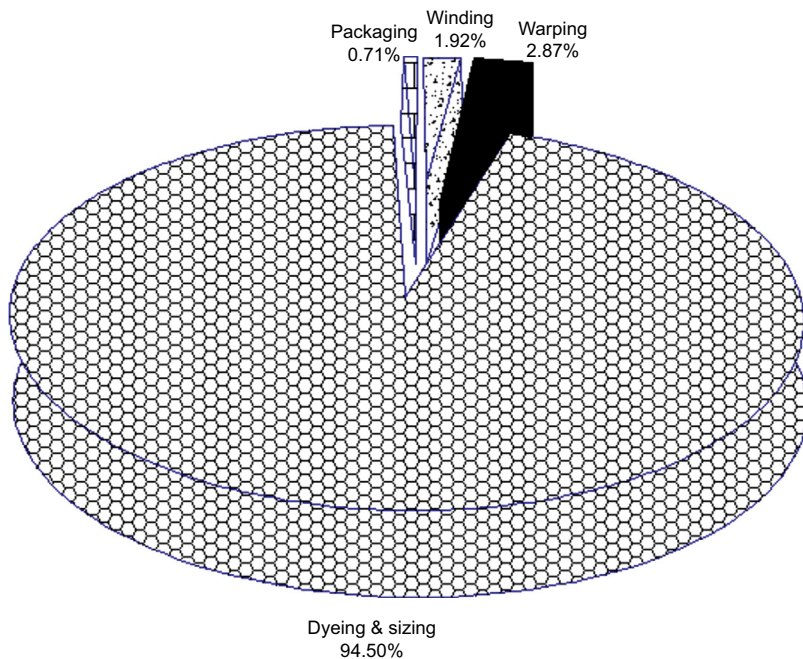


Figure 6.11 Greenhouse gas emissions in cotton dyeing, by process. CIAT, reproduced with permission. Clothing Industry Training Authority (CIAT), Apparel supply chain–carbon assessment about jeans producing. http://www.cita.org.hk/wp-content/uploads/2013/10/Fty_rpt_Eng_20130912.pdf.

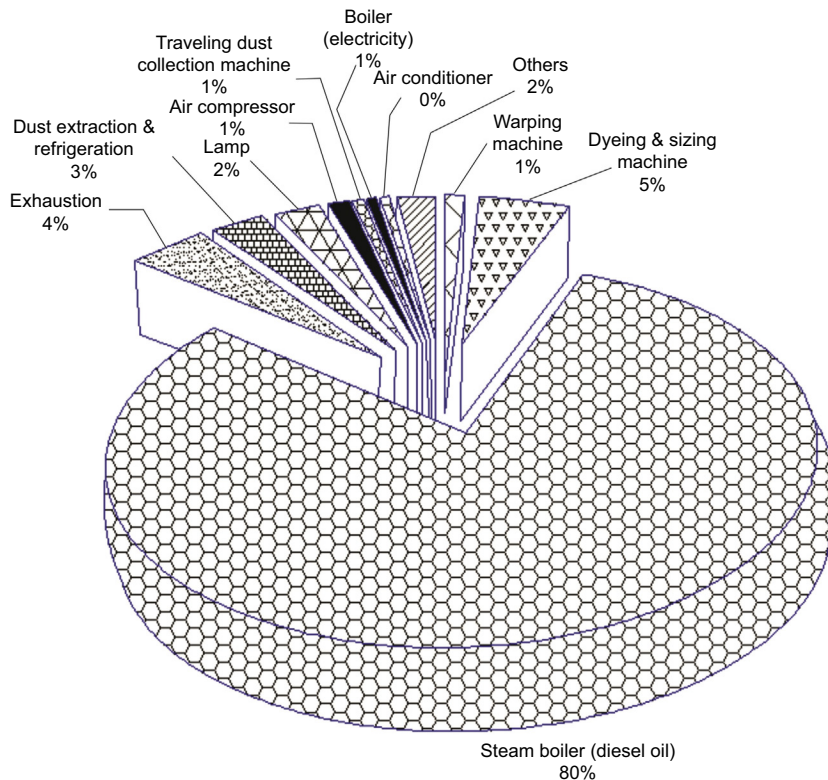


Figure 6.12 Greenhouse gas emission distribution in cotton dyeing, by emission source.

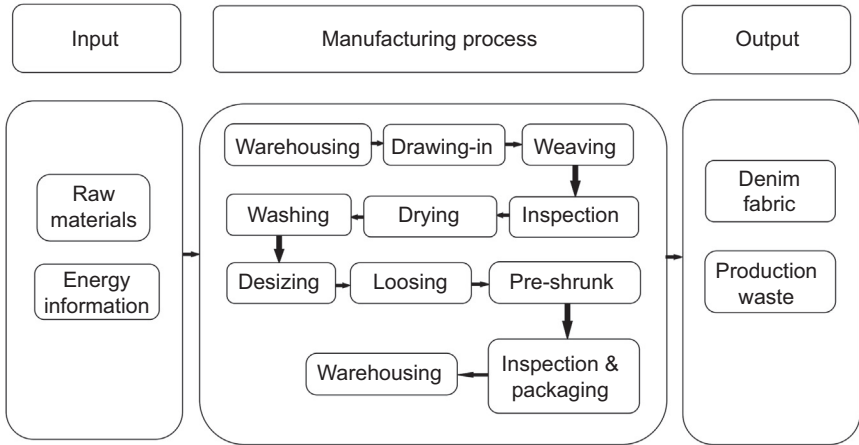


Figure 6.13 Process map of denim fabric weaving and finishing.

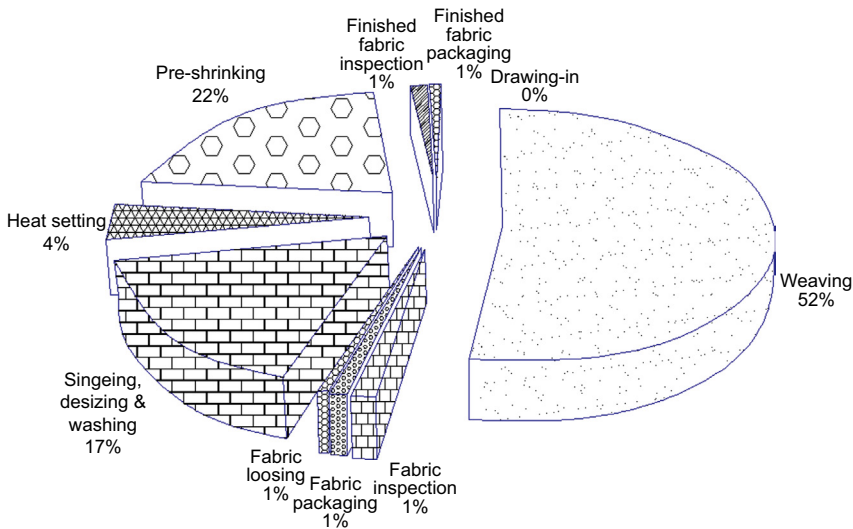


Figure 6.14 Greenhouse gas emissions in fabric weaving and finishing processes. CIAT, reproduced with permission. Clothing Industry Training Authority (CIAT), Apparel supply chain–carbon assessment about jeans producing. http://www.cita.org.hk/wp-content/uploads/2013/10/Fty_rpt_Eng_20130912.pdf.

The distribution of GHG emissions in denim fabric weaving and finishing, by process and emission source during the denim process, is shown in Figs 6.14 and 6.15, respectively. Fig. 6.14 shows that the weaving process alone contributes to about 50% of GHG emissions in the entire process. Fabric finishing processes such as preshrinking, singeing, desizing, washing and heat setting contribute around 40% of GHG emissions. Hence, weaving and finishing processes have to be thoroughly analyzed, including auxiliary production facilities, to reduce the carbon footprint [57,59].

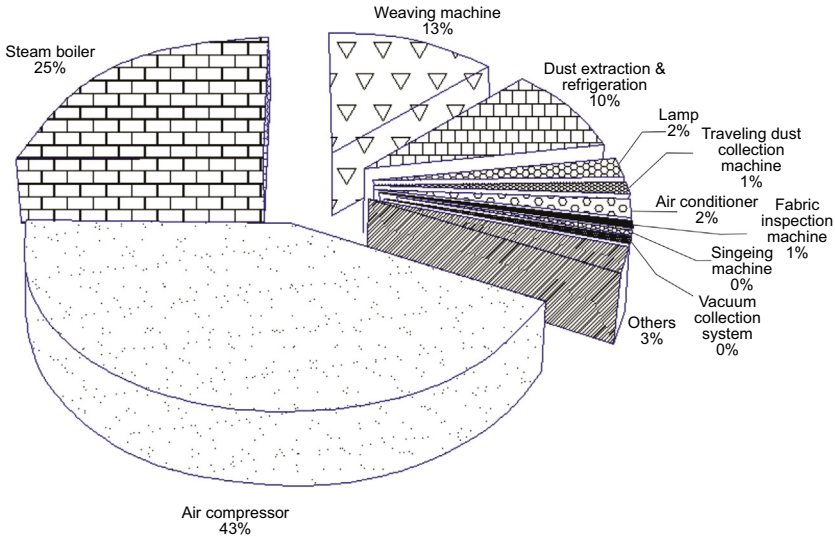


Figure 6.15 Greenhouse gas emission distribution in weaving and finishing, by emission source.

CIAT, reproduced with permission. Clothing Industry Training Authority (CIAT), Apparel supply chain–carbon assessment about jeans producing. http://www.cita.org.hk/wp-content/uploads/2013/10/Fty_rpt_Eng_20130912.pdf.

Fig. 6.15 shows that air compressors contribute to about 40% of total GHG emissions, followed by steam boilers (light oil) and weaving machines. Compressed air is an essential motive power in many processes of textile manufacturing. Air jet weaving needs a huge amount of compressed air. Because of the higher GHG emissions of air compressors (40%), they are considered an important issue for carbon footprint reduction [60]. The second major contribution is from steam boilers (20%) which are used to provide steam for finishing processes such as desizing, drying and preshrinking. Weaving machines combined with air compressors contribute the most GHG emissions in this sector.

6.6.5 Garment manufacturing

Garment manufacturing (otherwise known as the makeup process) is when two-dimensional denim fabrics are converted into three-dimensional garments. It involves pattern making, cutting, assembling and sewing, and finishing processes. The entire process map in denim garment manufacturing is shown in Fig. 6.16.

The distribution of GHG emissions in garment manufacturing, by process and emission source, is shown in Figs 6.17 and 6.18, respectively. Fig. 6.17 shows that the final ironing of garments before packing contributes about 58% of GHG emissions compared with other processes. The main garment manufacturing processes such as cutting and spreading account for about 17% of total GHG emissions. Ironing, which uses electric power and to remove creases and wrinkles, consumes a huge amount of energy [61–63]. The low production rate of the cutting process leads to higher GHG

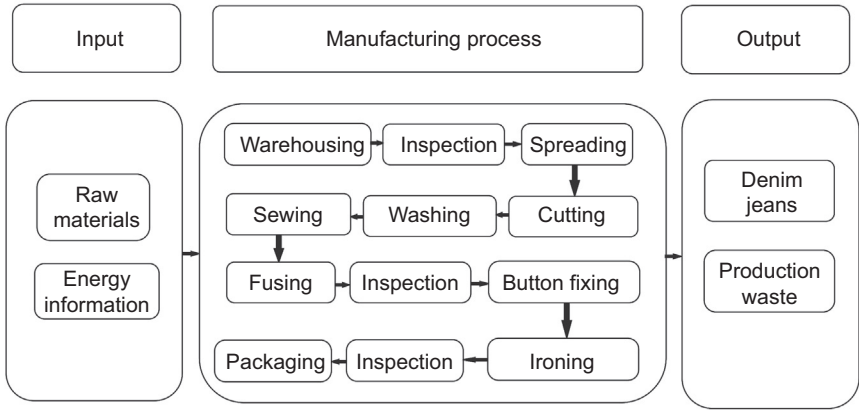


Figure 6.16 Process map in denim jeans manufacturing.

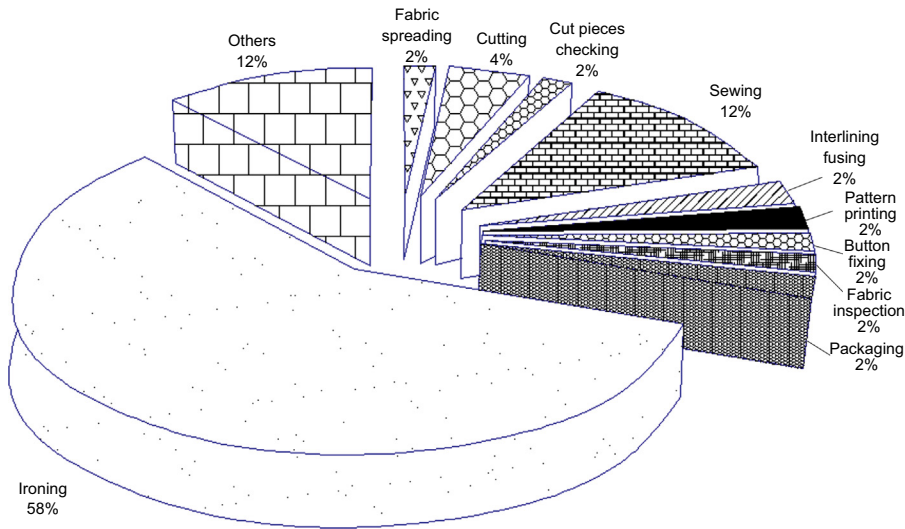


Figure 6.17 Greenhouse gas emissions in denim garment manufacturing process.

emissions. Fig. 6.18 shows that the steamer, lighting, sewing machine, exhaust fan, electric iron, air conditioner and fusing machines are major GHG emission sources and have to analyzed to reduce the carbon footprint.

6.6.6 Denim washing

To give jeans distinctive visual effects, a unique washing process is after the garment or fabric stage, such as a standard wash, stonewash, enzyme wash, chemical wash, bleach wash or destruction wash. The jeans-washing effect design is an important step of jeans development because it attracts consumers [64]. The washed products are sent back to the garment manufacturing factory for final inspection and packaging. This is

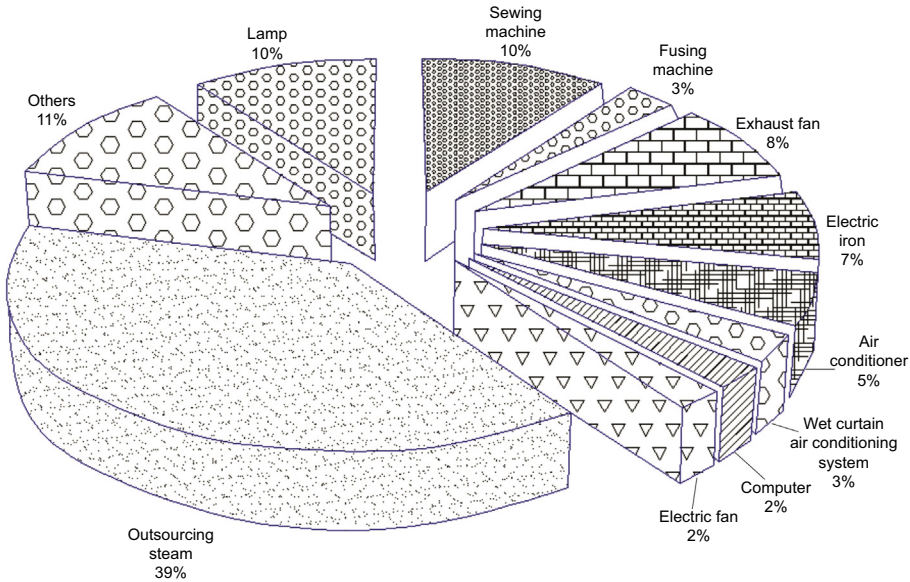


Figure 6.18 Greenhouse gas emission distribution in denim garment manufacturing, by emission source.

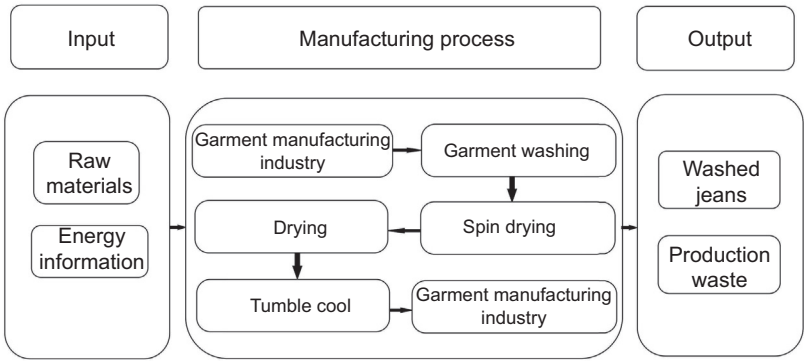


Figure 6.19 Process map in denim washing.

the end of the assessment of the carbon footprint for jean in the manufacturing sector. The entire process map in denim washing is shown in [Fig. 6.19](#).

The distribution of GHG emission in garment washing, by process and emission source, is shown in [Figs 6.20 and 6.21](#), respectively. [Fig. 6.20](#) shows that only four simple processes are involved in denim washing, of which the main washing process contributes more than 60% of total GHG emissions [65]. The second major contributor for GHG is a drying process. These two processes comprise the majority of GHG emissions with a total share of about 80%, which should be considered for opportunities to reduce carbon.

[Fig. 6.21](#) shows that outsourcing steam holds the biggest share of the total GHG emissions which are the essential elements for washing processes by providing

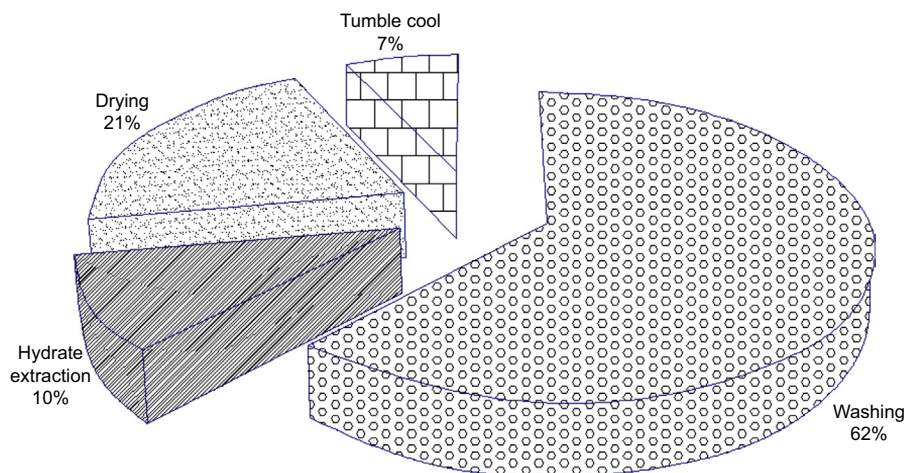


Figure 6.20 Greenhouse gas emissions in denim washing process.

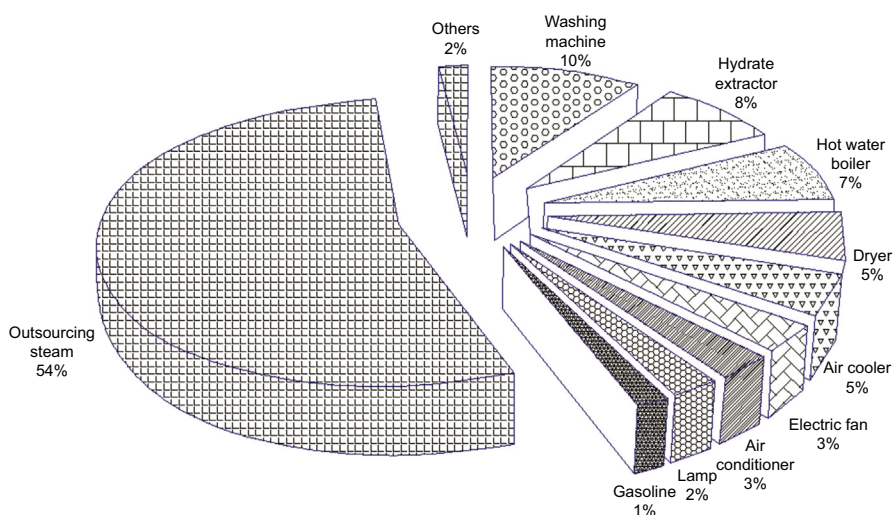


Figure 6.21 Greenhouse gas emission distribution in denim washing, by emission source.

steaming to enhance the washing effects. The electricity used to run washing machines contributes about 10% of GHG emissions.

6.6.7 Carbon footprint of denim in consumer use phase

The highest energy use and GHG emissions in the life cycle of denim come from the consumer use phase of jeans. Although overall water consumption in the jeans life cycle is accounted for at the cotton-growing stage, at around 68%, consumer care comes in at 23%. Based on these figures, it seems that the consumer use phase may have the greatest impact on the environment than any other phase. Fig. 6.22 shows

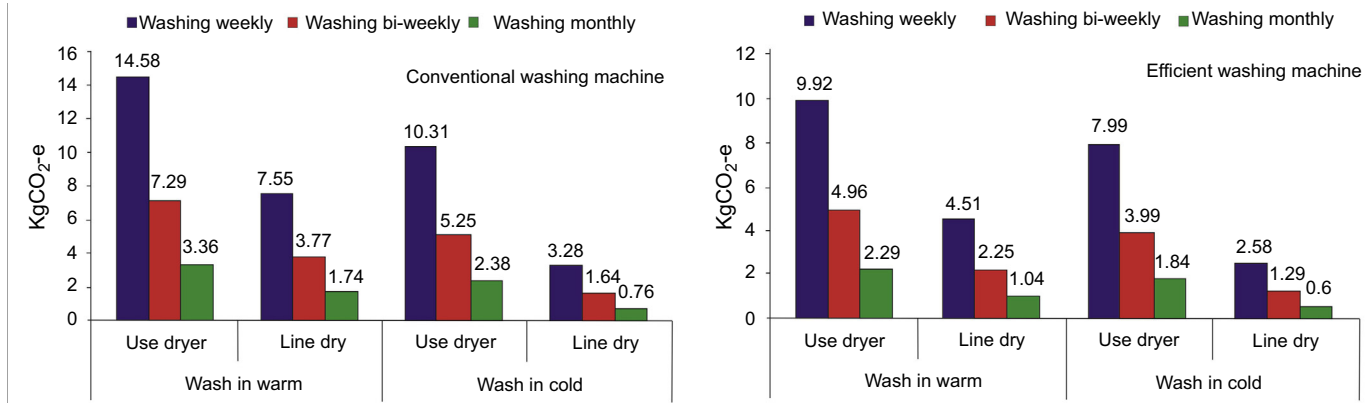


Figure 6.22 Impact of jeans on climate change from 1 year of consumer care.
Levi Strauss & Co. Life cycle of a jean, reproduced with permission.

that the overall impact on climate change is higher in the consumer use phase (37%), followed by fabric production (27%) [60].

The impact of a change in climate owing to consumers washing and drying jeans varies significantly based on the method of washing, washing frequency, and washing equipment. Fig. 6.22 shows the influence of the frequency of washing jeans, type of washing machine and washing conditions on the carbon footprint of jeans during the consumer use phase. From Fig. 6.22 it is evident that washing jeans with energy-efficient washing machines accounts for about a 30% lower carbon footprint compared with conventional washing machines; a further 30% reduction in carbon footprint was noticed by washing jeans in cold water compared with warm water. Similarly, washing jeans biweekly reduces the carbon footprint around 50% compared with washing them weekly, and a further reduction of 50% by washing them monthly [48]. Washing denim garments after they are worn 10 times instead of twice reduces energy use, the impact on climate change and water intake by up to 80%.

Fig. 6.23 shows the influence of washing machines and washing conditions on the carbon footprint on 1 day of washing jeans over various countries. Fig. 6.23 demonstrates that washing jeans in an energy-efficient washing machine and in cold water reduces carbon emissions compared with washing jeans in a conventional washing machine and in warm water [66,67]. Furthermore, the figure shows that consumers in China are leading the pack when it comes to laundering jeans, because they mostly wash in cold water and air-dry. American consumers have the highest water intake and use of nonrenewable energy. Consumers in the United Kingdom and France mostly air-dry their jeans but they use more hot water than do American or Chinese consumers. Consumers in the United States, United Kingdom and France wash their jeans more frequently than do consumers in China.

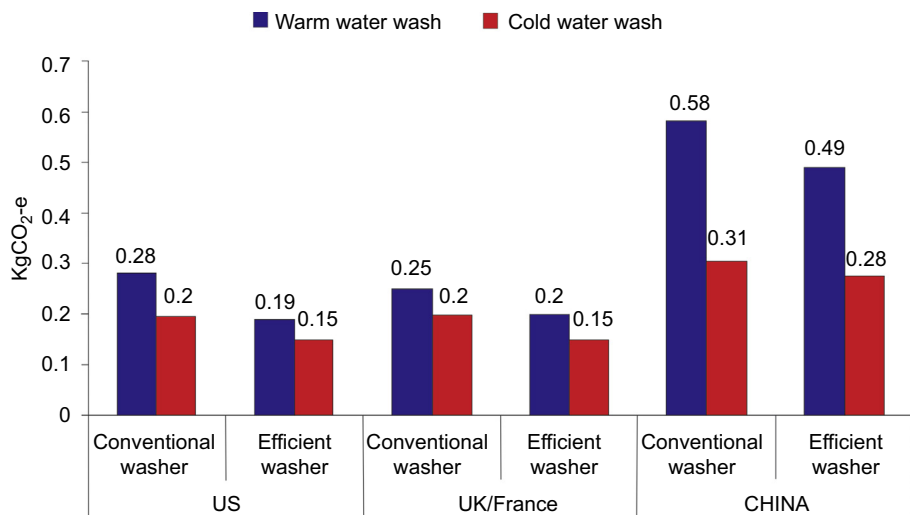


Figure 6.23 Global impact on climate change of 1 day of washing jeans. Levi Strauss & Co. Life cycle of a jean, reproduced with permission.

6.7 Strategies to reduce carbon footprint in denim manufacturing

A global paradigm shift toward cleaner and greener processes and products is in progress. A systematic approach with continuous improvement can minimize the carbon footprint of textiles. Textile industries have realized the benefits of increasing energy efficiency and thus reducing the costs of fuel and electricity. The following activities should be considered to reduce total life-cycle GHG emissions [68]:

- *Product design for the environment:* The decision regarding most significant life-cycle GHG emissions has to be done in the design stage of the product, so consideration of GHG and other environmental impacts should be incorporated into design decisions.
- *Care requirements:* Product care requirements and labels should be considered and consumers have to be educated so that garments are not washed more than necessary.
- *Enhancing product durability:* More hard-wearing products minimize GHG emissions substantially by spreading the 'sunk costs' of manufacturing emissions over a longer product life.
- *Review raw materials and engage with suppliers:* Consider greater use of plant-based fibres where garment specifications permit, engaging with suppliers to reduce emissions and shifting toward organic fibres to reduce GHG types.
- *Transport:* If air freight is used at any point in the supply chain, consider options to reduce it.

Environmental sustainability can be realized by considering the full life cycle of clothing, from the design and materials sourcing process onward to disposal. Total energy used in each stage of the process to produce the fabric is known as embodied energy and is the sum total of the energy required to produce the fibre and yarn as well as to weave those yarns into fabric. Beyond fibre production, dyeing and finishing are the highest consumers of water and energy in the textile processes and have the highest opportunities for energy and water savings and efficiency enhancements [69]. Once the magnitude of a carbon footprint is known for a product, a stratagem can be developed to reduce it by developing new technologies, an improved process and product management, changed Green Public or Private Procurement, consumption strategies, carbon capture, etc. Strategies that can be used to reduce the carbon footprint are:

- adopting the 3R concept (see subsequent discussion)
- developing standards and labels
- educating consumers

6.7.1 Adoption of 3R concept

6.7.1.1 Reduce

The carbon footprint in textiles can be reduced by processes and technologies which use relatively fewer raw materials and consume less energy, slowing down climate change.

Use of organic and wood-based fibres

To produce a single pair of jeans, about 3000 gallons of water is required, excluding chemicals, dyes, the energy required for drying and the workforce needed for production. Hence, there is a need to design sustainable practices and technology in the product supply chain. The battle is on to produce technology and processes that will lessen the impact of denim production on the environment as well as on the health of the workers making the items. To start with, more environmentally friendly fibres and dyes are needed which can use less water and safer chemicals [70].

By substituting traditional fibres with organic fibres and not using petrochemical-based fertilizers and pesticides during production, we can ensure a few GHGs are emitted. Some green fibres are bamboo, linen, soya bean, abaca, hemp and polyester recycled from used clothing. Besides having a low carbon footprint, natural fibres can be decomposed by microorganisms. Thus the fixed CO₂ in fibre is released and the cycle is closed. Also, natural fibres can sequester carbon, in which CO₂ from the atmosphere is absorbed by plants through photosynthesis and stored as carbon in biomass and soils [71].

Cultivation and organic cotton are inevitable for making sustainable denim. However, cultivation has to meet some standards that limit the use of toxic and agrochemicals. Hence, a more fastidious approach has to be practiced in organic cotton farming at all stages, including enhancing soil fertility, preharvest defoliation, and cotton picking. Growing organic cotton not only enhances natural ecosystems; it also increases human participation and a healthy lifestyle [24].

Replacing organic fibres for conventionally grown fibres is little better because it needs less energy for production and lower costs, and produces less CO₂. A study reported that organic fibres release 4.3% less GHG than conventional fibres and use only around 63% of energy for farming than conventional fibre farming. The important aspect is in handling organic matter; because soil organic matter is based on carbon, increasing the level of organic matters directly relates to carbon sequestration. Organic farming also supports environmental and social aspects and eliminates the use of synthetic fertilizers, pesticides which cause some serious problems. Because no pesticides are applied to organic cotton, organic cotton yields are as much as 50% lower than conventionally produced cotton, which necessitates more land to produce organic jeans. Organic agriculture is regarded as a climate change tool that can be used to battle global warming [72,73].

There is buzz in sustainable denim manufacturing about the use of denim from wood-based fibres. One eco-friendly fibre which can be used at a sustainable level to produce jeans is Tencel, made from wood pulp cellulose. The modal fibre is CO₂ neutral and can be combined with Tencel fibres to produce denim with stretch and performance. Some manufacturers use bamboo fibre, which is fast-growing and needs less water to cultivate. It does not require fertilizers and pesticides to manufacture sustainable jeans.

Reduction of water and chemicals in denim processing

Dye which can be employed in denim processing should have the property of reducing water use. Miguel Sanchez, global head of Lenzing product line for special dyes, says the firm is developing a dye that will reduce water usage by up to 92%. Most water consumed in colouring is in the washing-off stage (93%). Additional water is then

required to dry the yarn through steam (1%). In total, around 100,000L of water is used in 10,000m of yarn. Ideally water should go back to where it came from and close the loop. We really need to treat this water and send it back to nature [74]. The use of water for dyeing can be reduced by:

- using dyestuffs with 100% fixation capacity
- using 100% biodegradable dyestuffs and chemicals and ensuring the highest fixation possible in the application process
- minimizing washing-off
- recover and recycling everything
- replacing dye with another type of abundant solubilizing agent

Denim can be dyed using natural indigo to make it more sustainable. Besides the low yield of natural indigo production in plants, there is a problem because of the insolubility of indigo in water. Hence, the extraction of indigo from the natural plant requires a huge amount of alkali which contains a small amount of heavy metals. After the dyeing process most of the heavy metal can be seen in the blue jeans. The denim industry currently uses prereduced indigo crystals for dyeing, or a less sustainable powder indigo which has to be reduced at the mill using alkaline sodium hydrosulphite, the by-product of which is water. However the process poses a danger to human health because manual labour involved and there is exposure to waste [56].

The manufacturing and purification processes involved in producing synthetic indigo have a vital role in using of amounts of reducing agents and the quality of indigo dyes. The use of aniline as a starting material for the synthesis of indigo would not affect the environment because the quantity of it would be below critical limits. Based on the eco-efficiency analysis developed by BASF, it was shown that the total eco-balance of the production of blue jeans based on prereduced indigo and using an electrochemical dyeing (ECD) process would be optimal.

Energy-efficient processes

It is high time for the textile industry to implement energy-efficient and environmentally sound technologies to save costs and reduce the carbon footprint [56]:

- The ECD could replace the chemical reducing agent hydrosulphite completely with electrons for the dyeing of denim fabrics. This technique minimizes wastewater pollution considerably and enhances the ecological situation in the dye house. However, because of a labeling issue, the technology was not supported by retailers. This is the case with sulphur dyeing, especially sulphur black, the most important dyestuff for black denim. Currently, in sulphur dyeing process, the environmental friendly glucose is used as the best option as it causes huge chemical oxygen demand and Chemischen Sauerstoffbedarfs values in the effluent.
- After the completion of the yarn-dyeing and weaving process, the denim fabric is treated for special effects before going for making-up. The processing technologies which require a minimal amount of water, energy and chemicals in the process are needed for the eco-friendly processing of jeans. Innovative machines apply finishes to fabrics using foam, which conserves water.
- An ozone bleaching system can save up to 40% of energy and water and reduce cotton loss by 50%. For cleaner denim products, peroxide, oxygen and ozone bleach in garment finishing creates less environmental impact.

- Revolutionary air technology for dyeing requires only 25% of water compared with the conventional system and reduces energy and chemical consumption.
- Digital printing, using ink from the dyes, wastes neither fabric nor ink, does not use harmful salts and significantly reduces the environmental impact.
- Color-fast finish is a single-stage process for textiles and can reduce the processing time and CO₂ emissions.
- Industrial enzymes, which are basically proteins, replace strong chemicals used to remove impurities from the fibre or fabric; they reduce energy costs and water consumption and improve the feel of the fabric.

Effective laundry management

Effective laundering management in the context of sustainability means eliminating certain detergents and surfactants such as alkyl phenol and ethoxylates, and other chemicals such as phosphates and chlorine. Peracetic acid is considered a good alternative to chlorine bleach because it is less hazardous and more biodegradable.

Others

Setting up establishments that use more natural light for illumination, upgrading existing lighting systems with energy-efficient ones and installing heat recovery plants are all ways to reduce the carbon footprint.

6.7.1.2 Reuse

Effluents of chemically treated textiles are normally discharged in water. Treatment of wastewater obtained from chemically treated textiles is a must. Use of chrome mordant dyeing and limiting the emission of copper, chromium and nickel into water reduces impurities in dyes and pigments. Using dyeing carriers with high chlorine content should be avoided. During bleaching, an alternative agent can be used that is less hazardous or completely harmless. One challenge in the denim industry is the decolouring of wastewater. The concept involved in decolouring is zero liquid discharge. An electrochemical device is added to the water use area to treat the spent water before sludge is separated. Because no chemicals are used in the treatment, the process requires low energy with higher efficiency and reduced costs. DyStar introduced LoopTEC technology for wash water reuse and indigo recycling for denim dyeing; the company claimed 95% less water consumption and 95% less caustic demand, 95% less effluent volume and 75% energy savings.

6.7.1.3 Recycle

In denim manufacturing, the concept of reusing waste and raw materials is dominant in the production chain. Denim waste is being used for other application segments such as fleece and insulation material and to make denim bags. Denim recycling begins with using worn-out blue jeans as raw material. The pure cotton blue jeans are made free of all metal parts and the labels are shredded. After deinking, the shredded denim can be used in the paper industry. Also, the recycled cotton can be mixed with fresh cotton as a raw material in the production chain [69].

‘Real’ denim recycling employing used-up denim material as a raw material in the denim manufacturing was evaluated from only a few producers such as Orta Anadolu in Turkey. The denim sector is huge enough to attain remarkable technical and economical process if a number of denim brand manufacturers agree to the mutual concept. To implement this concept, only the absence of nonrecyclable raw materials is necessary. The denim is removed from other metal components and labels from the jeans by shredding. The shredded denim containing recycled cotton can be used as a raw material by mixing it with virgin cotton to manufacture denim. The recycled denim, called ‘R-Jeans’, could create an attractive marketing concept, especially for the growing sector of environmentally conscious customers.

Recycled cotton materials from used blue jeans are used as a part of carpet backing and sound absorption in Ford cars. Postconsumer cotton used in a car is obtained from approximately two pair of jeans. Another way to personalize old jeans is to recycle them into sandals to make soft, springy shoes; it makes a perfect accessory for both eco-warriors and funky fashionistas.

6.7.2 Development of standards and labels

6.7.2.1 The global recycling standard

This standard was developed to verify claims related to recycled goods. The percentage of recycling varies at 95%–100%, 70%–95% and 30%, respectively, for classifying them into gold, silver and bronze levels. Scientific certification systems have put forth criteria to be followed by ‘recycled’ materials. In addition to strict environmental processing, raw material specifications and social responsibility are included to ensure the health and safety of workers in accordance with International Labour Organization norms. The carbon trust, based in the United Kingdom, has developed the world’s first carbon label for clothing in association with continental clothing. This label provides the carbon footprint of garments right from raw materials and manufacturing, to use and disposal [24].

6.7.2.2 Global Organic Textile Standard

Internationally, the Global Organic Textile Standard (GOTS) is recognized as a tool for understanding environmentally sound production systems and social accountability in the textile sector. It covers the entire supply chain from raw material production to the distribution of all natural fibres. It actually means the use of certified organic fibres and the prohibition of an entire list of synthetic chemicals for use in the processes. Fabric that meets the norms of GOTS standards is more than just fabric; it offers a promise to keep the environment (air and water) pure and the soil renewed. Future generations will not be detrimentally affected by providing fabrics of GOTS standards, because of the reduced carbon footprint. The minimum requirement for certification from GOTS is that a product must be 67% recyclable or biodegradable.

6.7.3 Educate consumers

Consumer education regarding the huge carbon footprint in the textile value chain will inspire consumers to change their habits and attitudes. That is, consumers should minimize their purchase of fabrics that are blends of synthetic fibres and shift their focus toward 'organic fabrics'. In addition, they should be on the lookout for stamps from GOTS, SMART and Cradle to Cradle certifying agencies, paying attention to the carbon footprint of the fabrics purchased. On the other hand, retailers are also seeking sustainability requirements along their supply chains to maintain and expand their customer base [56,75].

Low-carbon manufacturing programmes and carbon accounting in factories, carbon footprint calculation projects, and benchmarking energy consumption across the textile and apparel supply chains are a few strategic measures required to reduce the carbon footprint of textile and apparel industry.

6.8 Conclusion

The textile industry across the globe is continuously taking steps toward reducing the carbon footprint to meet challenges for a more sustainable future. Awareness of environmental issues among consumers who insist on textile products that comply with environmental standards is also increasing. These are healthy trends that will continue to drive the industry toward offering eco-friendly products.

As a rule, in the future, innovative as well as eco-friendly processes and products should be economically attractive if plans related to environmental protection initiatives and the treatment of wastewater are carried out realistically. Most significant part in the ecological assessment and comparison of denim manufacturing concept is analyzing its features for the complete denim manufacturing process. Therefore only a total process comparison can offer a real comparison. The total sustainable denim production approach was highlight by DyStar, in its econfidence denim concept. The econfidence concept in the production of sustainable denim involves: ECD on slasher and rope dyeing ranges, use of minimal application processes in the denim manufacturing value chain, electrochemical bleaching in garment dyeing/finishing machines (laundry machines), combination processes and recycling concepts in denim laundries and garment finishing mills, and eco-efficiency evaluation of denim processes.

Sustainable denim manufacturing in the near future will be a process involving all steps of the complete denim manufacturing value chain. The optimal solution comes from balancing the use of recycled materials, advanced and energy-efficient machinery and equipment for manufacturing and use of optimized process parameters in manufacturing. In this process of development it would be helpful if retailers (global jeans brands) implement a global jeans recycling concept and use eco-labeling on best available technology parameters. A reduction in the global warming potential of denim manufacturing and production can come from using minimal resources, water, chemicals and energy as well as a minimal carbon footprint.

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